A. PROOF OF LEMMA 1

Let us demonstrate the validity of Lemma 1. Consider the stochastic gradient vector $\widehat{\boldsymbol{G}}_i$ defined with the τth block being

$$[\widehat{\boldsymbol{G}}_i]_\tau = \mathrm{col}\left\{\widehat{\boldsymbol{g}}_k^{(\tau)}(\boldsymbol{x}_{k,i-1})\right\}_{k\in\mathcal{N}}. \tag{27}$$

First, we show that the m-ATC-ITC algorithm can be written as

$$\boldsymbol{X}_i = \mathcal{P}\left(\boldsymbol{X}_{i-1} - \mathcal{M}\widehat{\boldsymbol{G}}_i\right). \tag{28}$$

Notice that the matrix $\mathcal{P}$ is block-diagonal by definition. We separate the analysis for each $\tau = \tau' \in [T]$. For τ fixed, consider $t = \tau$,

$$\begin{aligned}
[\boldsymbol{X}_i]_{\tau,t} &= \sum_{t'\in[T]} [\mathcal{P}]_{\tau\tau,tt'}\left([\boldsymbol{X}_{i-1}]_{\tau,t'} - [\mathcal{M}\widehat{\boldsymbol{G}}_i]_{\tau,t'}\right) \\
&= \sum_{t'\in[T]} \left\{\begin{array}{ll} A^{(\tau)\mathsf{T}} \otimes I_{M_\tau}, & \tau = t = t'; \\ 0_{K_t M_\tau \times K_{t'} M_\tau}, & \text{otherwise} \end{array}\right\}. \\
&\quad \left(\mathrm{col}\left\{\boldsymbol{x}_{k,i-1}^{(\tau)} - \mu^{(\tau)}\widehat{\boldsymbol{g}}_{k,i}^{(\tau)}\right\}_{k\in\mathcal{N}^{(t')}}\right) \\
&= \left(A^{(\tau)\mathsf{T}} \otimes I_{M_\tau}\right) \cdot \\
&\quad \mathrm{col}\left\{\boldsymbol{x}_{k,i-1}^{(\tau)} - \mu^{(\tau)}\widehat{\boldsymbol{g}}_{k,i}^{(\tau)}\right\}_{k\in\mathcal{N}^{(t)}} \\
&= \mathrm{vec}\left(\mathrm{row}\left\{\boldsymbol{x}_{k,i-1}^{(\tau)} - \mu^{(\tau)}\widehat{\boldsymbol{g}}_{k,i}^{(\tau)}\right\}_{k\in\mathcal{N}^{(t)}} A^{(\tau)}\right).
\end{aligned}$$

This is the ATC update. The last equality is by the vectorization identity

$$\mathrm{vec}(AXB) = \left(B^{\mathsf{T}} \otimes A\right)\mathrm{vec}(X). \tag{29}$$

Next, consider $t \neq \tau$,

$$\begin{aligned}
[\boldsymbol{X}_i]_{\tau,t} &= \sum_{t'\in[T]} [\mathcal{P}]_{\tau\tau,tt'}\left([\boldsymbol{X}_{i-1}]_{\tau,t'} - [\mathcal{M}\widehat{\boldsymbol{G}}_i]_{\tau,t'}\right) \\
&= \sum_{t'\in[T]} \left\{\begin{array}{ll} C^{(tt)\mathsf{T}} \otimes I_{M_\tau}, & \tau \neq t = t'; \\ (A^{(\tau)}C^{(\tau t)})^{\mathsf{T}} \otimes I_{M_\tau}, & \tau = t' \neq t; \\ 0_{K_t M_\tau \times K_{t'} M_\tau}, & \text{otherwise} \end{array}\right\}. \\
&\quad \left(\mathrm{col}\left\{\boldsymbol{x}_{k,i-1}^{(\tau)} - \mu^{(\tau)}\widehat{\boldsymbol{g}}_{k,i}^{(\tau)}\right\}_{k\in\mathcal{N}^{(t')}}\right) \\
&= \left(C^{(tt)\mathsf{T}} \otimes I_{M_\tau}\right) \cdot \\
&\quad \mathrm{col}\left\{\boldsymbol{x}_{k,i-1}^{(\tau)} - \mu^{(\tau)}\widehat{\boldsymbol{g}}_{k,i}^{(\tau)}\right\}_{k\in\mathcal{N}^{(t)}} + \\
&\quad \left((A^{(\tau)}C^{(\tau t)})^{\mathsf{T}} \otimes I_{M_\tau}\right) \cdot \\
&\quad \mathrm{col}\left\{\boldsymbol{x}_{k,i-1}^{(\tau)} - \mu^{(\tau)}\widehat{\boldsymbol{g}}_{k,i}^{(\tau)}\right\}_{k\in\mathcal{N}^{(\tau)}}.
\end{aligned}$$

The first term above is the sharing of estimates within team t, the second term above is equal to

$$\left(C^{(\tau t)})^{\mathsf{T}} \otimes I_{M_\tau}\right) \cdot [\boldsymbol{X}_i]_{\tau,\tau},$$

which represents information flowing from team τ to team t. The two terms combined hence represents the ITC update.

Second, notice that by the left-stochasticity of $A^{(\tau)}$ and $C^{(\tau t)}$, we have

$$\begin{aligned}
[\mathcal{P}X^\star]_{\tau,t} &= \sum_{\tau',t'\in[T]} [\mathcal{P}]_{\tau\tau',tt'}[X^\star]_{\tau',t'} \\
&= \sum_{\tau',t'\in[T]} [\mathcal{P}]_{\tau\tau',tt'}\left(\mathbf{1}_{K_{t'}} \otimes x^{(\tau'),\star}\right) \\
&= \mathbf{1}_{K_t} \otimes x^{(\tau),\star} = [X^\star]_{\tau,t},
\end{aligned}$$

viz., $\mathcal{P}X^\star = X^\star$. Subtract $X^\star$ by Eqn. (28) to obtain

$$\widetilde{\boldsymbol{X}}_i = \mathcal{P}\left(\widetilde{\boldsymbol{X}}_{i-1} + \mathcal{M}\widehat{\boldsymbol{G}}_i\right). \tag{30}$$

Third, we separate the gradient noise from the stochastic gradient. I.e., by defining the expected gradient conditioning on the filtration $\boldsymbol{\mathcal{F}}_{i-1}$ as $\boldsymbol{G}_i$, its τth block is

$$[\boldsymbol{G}_i]_\tau = \mathrm{col}\left\{\boldsymbol{g}_k^{(\tau)}(\boldsymbol{x}_{k,i-1})\right\}_{k\in\mathcal{N}}. \tag{31}$$

Then, Eqn. (30) becomes

$$\widetilde{\boldsymbol{X}}_i = \mathcal{P}\left(\widetilde{\boldsymbol{X}}_{i-1} + \mathcal{M}\boldsymbol{G}_i\right) + \mathcal{P}\mathcal{M}\boldsymbol{S}_i, \tag{32}$$

with $\boldsymbol{S}_i = \widehat{\boldsymbol{G}}_i - \boldsymbol{G}_i$.

Lastly, we will show that $\boldsymbol{G}_i = G^\star - \boldsymbol{\mathcal{H}}_{i-1}\widetilde{\boldsymbol{X}}_{i-1}$.

Lemma 2. *All the gradients can be written in the form of*

$$\nabla_\tau J_{\ell_k}^{(\tau)}(\boldsymbol{x}_{k,i}) = \nabla_\tau J_{\ell_k}^{(\tau)}(x^\star) - \sum_{t\in[T]} \boldsymbol{H}_{k,\tau t,i}\widetilde{\boldsymbol{x}}_{k,i}^{(t)}. \tag{33}$$

where $\widetilde{\boldsymbol{x}}_{k,i}^{(t)} := x^{(t),\star} - \boldsymbol{x}_{k,i}^{(t)}$ *and* $\boldsymbol{H}_{k,\tau t,i}$ *is an integral of* $\nabla_{t\tau}^2 J_{\ell_k}^{(\tau)}$.

Proof. The lemma is a direct result of the fundamental theorem of calculus. The resulting $\boldsymbol{H}_{k,\tau t,i}$'s are not unique, but the non-uniqueness does not affect the validity of the proofs. The following choice is made so that the forthcoming derivations are cleaner. Since

$$\begin{aligned}
&\nabla_\tau J_{\ell_k}^{(\tau)}(x^\star) - \nabla_\tau J_{\ell_k}^{(\tau)}(\boldsymbol{x}_{k,i}) \\
&= \underbrace{\int_0^1 \nabla\left(\nabla_\tau J_{\ell_k}^{(\tau)}(x^\star - \eta\widetilde{\boldsymbol{x}}_{k,i})\right)\mathrm{d}\eta}_{\left[\boldsymbol{H}_{k,\tau 1,i} \quad \cdots \quad \boldsymbol{H}_{k,\tau T,i}\right]}\widetilde{\boldsymbol{x}}_{k,i},
\end{aligned}$$

the integral is partitioned into matrices $\boldsymbol{H}_{k,\tau t,i}$ of size $M_\tau \times M_t$, with each block having the explicit form

$$\boldsymbol{H}_{k,\tau t,i} = \int_0^1 \nabla_{t\tau} J_{\ell_k}^{(\tau)}(x^\star - \eta\widetilde{\boldsymbol{x}}_{k,i})\,\mathrm{d}\eta.$$

□

Continue on with the definitions established in Lemma 1, by defining

$$\boldsymbol{\mathcal{H}}^{(t)}_{\tau\tau',i-1} := \text{blkdiag}\{\boldsymbol{H}_{k,\tau\tau',i-1}\}_{k\in\mathcal{N}^{(t)}}, \tag{34}$$

we have

$$\begin{aligned}
[\boldsymbol{G}_i]_{\tau,t} &= \text{col}\left\{\nabla_\tau J^{(\tau)}_{\ell_k}(\boldsymbol{x}_{k,i-1})\right\}_{k\in\mathcal{N}^{(t)}} \\
&= \text{col}\left\{\nabla_\tau J^{(\tau)}_{\ell_k}(x^\star)\right\}_{k\in\mathcal{N}^{(t)}} - \\
&\quad \text{col}\left\{\sum_{\tau'\in[T]} \boldsymbol{H}_{k,\tau\tau',i-1}\widetilde{\boldsymbol{x}}^{(\tau')}_{k,i-1}\right\}_{k\in\mathcal{N}^{(t)}} \\
&= [G^\star]_{\tau,t} - \sum_{\tau'\in[T]} \text{col}\left\{\boldsymbol{H}_{k,\tau\tau',i-1}\widetilde{\boldsymbol{x}}^{(\tau')}_{k,i-1}\right\}_{k\in\mathcal{N}^{(t)}} \\
&= [G^\star]_{\tau,t} - \sum_{\tau'\in[T]} \text{blkdiag}\left\{\boldsymbol{H}_{k,\tau\tau',i-1}\right\}_{k\in\mathcal{N}^{(t)}} \cdot \\
&\quad \text{col}\left\{\widetilde{\boldsymbol{x}}^{(\tau')}_{k,i-1}\right\}_{k\in\mathcal{N}^{(t)}} \\
&= [G^\star]_{\tau,t} - \sum_{\tau'\in[T]} [\boldsymbol{\mathcal{H}}_{i-1}]_{\tau\tau',tt}[\widetilde{\boldsymbol{X}}_{i-1}]_{\tau',t} \\
&= [G^\star]_{\tau,t} - \sum_{\tau',t'\in[T]} [\boldsymbol{\mathcal{H}}_{i-1}]_{\tau\tau',tt'}[\widetilde{\boldsymbol{X}}_{i-1}]_{\tau',t'}.
\end{aligned}$$

Thus, it is shown that $\boldsymbol{G}_i = G^\star - \boldsymbol{\mathcal{H}}_{i-1}\widetilde{\boldsymbol{X}}_{i-1}$. Combining the decomposition of $\boldsymbol{G}_i$ established above and Eqn. (32), we finally obtain Eqn. (20) of Lemma 1.

Finally, it should be noted that the operations col, row, vec, and blkdiag are used in accordance to [1, Appendices].

B. ANCILLARY LEMMAS FOR SECOND MOMENT STABILITY

The general proof to the stability of the m-ATC-ITC algorithm is very much similar to the techniques developed by Sayed et al. [1] for diffusion learning over network. However, the multi-team network structure requires some adaptation of some key lemmas and theorems.

This section of the appendix provide the necessary lemmas for the proof of the second moment stability present in the next section. In summary, Lemma 3 to Lemma 5 are used to derive Theorem 2. In Theorem 2, we transform the error recursion of Eqn. (20) into the Perron coordinates of Eqn. (39). Then, Lemma 7 to Lemma 11 are is used to bound the terms in Eqn. (39), which is illustrated in later sections.

Lemma 3. *Under Assumption 1, the matrix $[\mathcal{P}]_{\tau\tau}$ is left-stochastic and has a single eigenvalue 1 with multiplicity M_τ on the spectral circle for all $\tau \in [T]$.*

Proof. It is trivial to demonstrate the left-stochasticity. We have also used this property when deriving $\mathcal{P}X^\star = X^\star$. This implies the spectral radius of $[\mathcal{P}]_{\tau\tau}$ is 1. It remains to show that only one eigenvalue exists on the spectral circle.

By a permutation of the rows and columns, the block $[\mathcal{P}^\mathsf{T}]_{\tau\tau}$ is similar to

$$\left[\begin{array}{c|ccc} A^{(\tau)} & A^{(\tau)}C^{(\tau 1)} & \cdots & A^{(\tau)}C^{(\tau T)} \\ \hline & C^{(11)} & & \\ & & \ddots & \\ & & & C^{(TT)} \end{array}\right] \otimes I_{M_\tau}.$$

The spectrum of $[\mathcal{P}]_{\tau\tau}$ is hence equal to the union of the spectra of $A^{(\tau)}$ and $C^{(tt)}$'s (with $t \neq \tau$) with multiplicity M_τ.

Since $A^{(\tau)}$ is primitive, it has a simple (single) eigenvalue 1 on its spectral circle. Further, since $C^{(\tau t)}$ is non-zero for all $t \neq \tau$, the columns to $C^{(tt)}$ adds up to a value less than or equal to 1, and at least one column has a strict inequality. It can hence be shown that the spectral radius $\rho(C^{(tt)}) < 1$. We prove it by contradiction: Consider an eigenvalue and eigenvector pair (λ, x) to $C^{(tt)}$ where $|\lambda| = 1$. Define the element-wise absolute value to x as $y := |x|$. Consider the inequality signs being used in an element-wise fashion, we have

$$C^{(tt)}y = \left|C^{(tt)}\right||x| \geq \left|C^{(tt)}x\right| = |\lambda x| = |\lambda||x| = y$$

by the triangle inequality. Summing up all the elements, we have

$$\mathbf{1}^\mathsf{T}y \leq \mathbf{1}^\mathsf{T}C^{(tt)}y < \mathbf{1}^\mathsf{T}y.$$

The strict inequality is due to there being at least a column to $C^{(tt)}$ with sum strictly less than 1. A contradiction is thus met. All eigenvalues to $C^{(tt)}$ have magnitude strictly less than 1. Thus, we have proven the statement. □

Lemma 4 (Eigenvectors of $\mathcal{P}$)**.** *Suppose $p^{(\tau)}$ is the Perron vector of $A^{(\tau)}$. Let $K_{1:\tau-1} = K_1 + \cdots + K_{\tau-1}$ and $K_{\tau+1:T} = K_{\tau+1} + \cdots + K_T$. The matrices $\mathfrak{p}^{(\tau)}$ and $\mathfrak{q}^{(\tau)} \in \mathbb{R}^{KM_\tau \times M_\tau}$ defined as*

$$\mathfrak{p}^{(\tau)} := \text{col}\left\{0_{K_{1:\tau-1}\times 1}, p^{(\tau)}, 0_{K_{\tau+1:T}\times 1}\right\} \otimes I_{M_\tau}, \tag{35a}$$

$$\mathfrak{q}^{(\tau)} := \mathbf{1}_K \otimes I_{M_\tau} \tag{35b}$$

satisfies

$$\mathfrak{p}^{(\tau)\mathsf{T}}[\mathcal{P}]_{\tau\tau} = \mathfrak{p}^{(\tau)\mathsf{T}}, \qquad [\mathcal{P}]_{\tau\tau}\mathfrak{q}^{(\tau)} = \mathfrak{q}^{(\tau)}. \tag{36}$$

Proof. The proof is direct by calculating

$$\left[\mathfrak{p}^{(\tau)\mathsf{T}}[\mathcal{P}]_{\tau\tau}\right]_t = \sum_{t'\in[T]} [\mathfrak{p}^{(\tau)}]^\mathsf{T}_{t'}[\mathcal{P}]_{\tau\tau,t't}$$

and

$$\left[[\mathcal{P}]_{\tau\tau}\mathfrak{q}^{(\tau)}\right]_t = \sum_{t'\in[T]} [\mathcal{P}]_{\tau\tau,tt'}[\mathfrak{q}^{(\tau)}]_{t'}.$$

The detailed calculations are similar to that encountered in the proof of Lemma 1 and are thus omitted. □

Note that the weighting coefficients $p_k^{(t)}$'s in Eqn. (1a) are exactly the entries to $p^{(t)}$.

From Lemma 4, we can perform a spectral decomposition on $\mathcal{P}$, and consequently, transform Eqn. (20) into its Perron coordinates. This is a natural transformation to perform since the components in the Perron subspace represent the consensus of the network. The results on the convergence of a signal essentially amount to bounding the behaviors of the out-of-consensus components of the signal.

The exact form of the coordinate transformation is described in the lemma below.

Lemma 5 (Spectral decomposition of $\mathcal{P}$)**.** *For all $\tau \in [T]$, denote a Jordan form of $[\mathcal{P}]_{\tau\tau}$ as*

$$\left[\begin{array}{c|cccc} 1 & & & & \\ \hline & \lambda_{2,\tau} & \varepsilon_\tau & & \\ & & \lambda_{2,\tau} & \ddots & \\ & & & \ddots & \\ & & & & \end{array}\right] \otimes I_{M_\tau} = \left[\begin{array}{c|c} 1 & \\ \hline & J_{\varepsilon_\tau} \end{array}\right] \otimes I_{M_\tau},$$

where the first upper-diagonal elements are scaled to ε_τ. The choice of ε_τ's is discussed in Appendix C. Then, there exists a complex matrix $\mathcal{V}$ of size $KM \times KM$ such that

$$\mathcal{P} = \mathcal{V}^{-\mathsf{H}} \mathcal{J} \mathcal{V}^{\mathsf{H}}, \tag{37}$$

with

$$\mathcal{J} = \begin{bmatrix} \mathcal{J}_{11} & 0 \\ 0 & \mathcal{J}_{22} \end{bmatrix}, \tag{38a}$$

$$\mathcal{J}_{11} = I_M, \tag{38b}$$

$$\mathcal{J}_{22} = \operatorname{blkdiag}\left\{J_{\varepsilon_\tau} \otimes I_{M_\tau}\right\}_{\tau \in [T]}. \tag{38c}$$

Proof. Since

$$\mathfrak{p}^{(\tau)\mathsf{T}} \mathfrak{q}^{(\tau)} = I_{M_\tau},$$

let the Jordan decomposition of $[\mathcal{P}]_{\tau\tau}$ be

$$[\mathcal{P}]_{\tau\tau} = \left[\begin{array}{c|c} \mathfrak{q}^{(\tau)} & V_{L,\tau} \end{array}\right] \left[\begin{array}{c|c} I_{M_\tau} & \\ \hline & J_{\varepsilon_\tau} \otimes I_{M_\tau} \end{array}\right] \left[\begin{array}{c} \mathfrak{p}^{(\tau)\mathsf{T}} \\ \hline V_{R,\tau}^{\mathsf{H}} \end{array}\right],$$

where

$$V_{R,\tau}^{\mathsf{H}} \mathfrak{q}^{(\tau)} = 0_{(K-1)M_\tau \times M_\tau},$$
$$\mathfrak{p}^{(\tau)\mathsf{T}} V_{L,\tau} = 0_{M_\tau \times (K-1)M_\tau},$$
$$V_{R,\tau}^{\mathsf{H}} V_{L,\tau} = I_{(K-1)M_\tau}.$$

Then, for $\tau \in [T]$ and $m \in [2T]$, define the matrix

$$[\mathcal{V}]_{\tau m} := \begin{cases} \mathfrak{p}^{(m)}, & m \le T, \tau = m; \\ 0_{KM_\tau \times KM_m}, & m \le T, \tau \ne m; \\ V_{R,m-T}, & m > T, \tau = m - T; \\ 0_{KM_\tau \times (K-1)M_{m-T}}, & m > T, \tau \ne m - T \end{cases}$$

and the inverse of its Hermitian conjugate

$$[\mathcal{V}^{-\mathsf{H}}]_{\tau m} := \begin{cases} \mathfrak{q}^{(m)}, & m \le T, \tau = m; \\ 0_{KM_\tau \times KM_m}, & m \le T, \tau \ne m; \\ V_{L,m-T}, & m > T, \tau = m - T; \\ 0_{KM_\tau \times (K-1)M_{m-T}}, & m > T, \tau \ne m - T. \end{cases}$$

The result is immediately shown by plugging $\mathcal{V}$ and $\mathcal{V}^{-\mathsf{H}}$ specified above into Eqn. (37). □

Remark that so far we have used three types of indexing: $\tau, \tau' \in [T]$ for the indexing of larger blocks, each of size $KM_\tau \times KM_{\tau'}$; $t, t' \in [T]$ for the indexing in each $(\tau\tau')$-block, each of size $K_t M_\tau \times K_{t'} M_{\tau'}$; and now the new index $m \in [2T]$ which replaces the indices $(\tau\tau')$, representing the blocks in a new set of coordinates. The block matrices in this new set of coordinates have sizes following those illustrated in the lemma above.

The following is an example of $T = 3$ with the index (τm) of each block of $\mathcal{V}$ marked:

$$\left[\begin{array}{ccc|ccc} 11 & 12 & 13 & 14 & 15 & 16 \\ 21 & 22 & 23 & 24 & 25 & 26 \\ 31 & 32 & 33 & 34 & 35 & 36 \end{array}\right].$$

It will be convenient for us to partition $\mathcal{V}$ and $\mathcal{V}^{-\mathsf{H}}$ into blocks of size $KM \times M$ and $KM \times (K-1)M$ as shown by the dashed line above. We denote the partitioned blocks as

$$\mathcal{V} = \left[\begin{array}{c|c} \mathfrak{p} & \mathcal{V}_R \end{array}\right],$$
$$\mathcal{V}^{-\mathsf{H}} = \left[\begin{array}{c|c} \mathfrak{q} & \mathcal{V}_L \end{array}\right].$$

Theorem 2 (Coordinate transform on error recursion)**.** *Let $\widetilde{\boldsymbol{W}}_i := \mathcal{V}^{\mathsf{H}} \widetilde{\boldsymbol{X}}_i$, the error recursion of Eqn. (20) can be rewritten in the form*

$$\widetilde{\boldsymbol{W}}_i = \left(\mathcal{J} - \overline{\mathcal{M}} \boldsymbol{\mathcal{D}}_{i-1}\right) \widetilde{\boldsymbol{W}}_{i-1} + \overline{\mathcal{M}} \mathcal{G}^\star + \overline{\mathcal{M}} \boldsymbol{\mathcal{S}}_i, \tag{39}$$

where the matrices admit the following partition:

$$\widetilde{\boldsymbol{W}}_i = \operatorname{col}\left\{\widetilde{\boldsymbol{W}}_{1,i}, \widetilde{\boldsymbol{W}}_{2,i}\right\}, \tag{40a}$$

$$\mathcal{J} = \operatorname{blkdiag}\{\mathcal{J}_{11}, \mathcal{J}_{22}\}, \tag{40b}$$

$$\overline{\mathcal{M}} = \operatorname{blkdiag}\left\{\mathcal{M}_{11}, \mathcal{M}_{22}\right\}, \tag{40c}$$

$$\boldsymbol{\mathcal{D}}_i = \begin{bmatrix} \boldsymbol{\mathcal{D}}_{11,i} & \boldsymbol{\mathcal{D}}_{12,i} \\ \boldsymbol{\mathcal{D}}_{21,i} & \boldsymbol{\mathcal{D}}_{22,i} \end{bmatrix}, \tag{40d}$$

$$\mathcal{G}^\star = \operatorname{col}\left\{0_{M\times 1}, \mathcal{G}_2^\star\right\}, \tag{40e}$$

$$\boldsymbol{\mathcal{S}}_i = \operatorname{col}\left\{\boldsymbol{\mathcal{S}}_{1,i}, \boldsymbol{\mathcal{S}}_{2,i}\right\}. \tag{40f}$$

The partition is consistent across the matrices.

The size of each block matrix is consistent with the sizes of the blocks in $\mathcal{D}_i$: to be more specific, we have $\boldsymbol{\mathcal{D}}_{11,i} \in \mathbb{R}^{M\times M}$, $\boldsymbol{\mathcal{D}}_{12,i} \in \mathbb{C}^{M\times(K-1)M}$, $\boldsymbol{\mathcal{D}}_{21,i} \in \mathbb{C}^{(K-1)M\times M}$, and $\boldsymbol{\mathcal{D}}_{22,i} \in \mathbb{C}^{(K-1)M\times(K-1)M}$.

Proof. Since $\mathcal{J}$ is already defined, we only need to specify $\overline{\mathcal{M}}$, $\boldsymbol{\mathcal{D}}_i$, $\mathcal{G}^\star$, and $\boldsymbol{\mathcal{S}}_i$. First, define

$$\overline{\mathcal{M}} := \mathcal{V}^{\mathsf{H}} \mathcal{M} \mathcal{V}^{-\mathsf{H}},$$

we have

$$\mathcal{M}_{11} = \text{blkdiag}\left\{\mu^{(\tau)} I_{M_\tau}\right\}_{\tau\in[T]},$$
$$\mathcal{M}_{22} = \text{blkdiag}\left\{\mu^{(\tau)} I_{(K-1)M_\tau}\right\}_{\tau\in[T]}.$$

Second, define

$$\boldsymbol{\mathcal{D}}_i := \mathcal{V}^{\mathsf{H}} \boldsymbol{\mathcal{H}}_i \mathcal{V}^{-\mathsf{H}} = \begin{bmatrix} \mathfrak{p}^{\mathsf{T}} \\ \mathcal{V}_R^{\mathsf{H}} \end{bmatrix} \boldsymbol{\mathcal{H}}_i \begin{bmatrix} \mathfrak{q} & \mathcal{V}_L \end{bmatrix},$$

the four blocks are analyzed separately. The upper-left block has the following closed form expression: for $m, n \in [T]$, since

$$\begin{aligned}
[\boldsymbol{\mathcal{D}}_{11,i}]_{mn} &= [\mathfrak{p}^{\mathsf{T}} \boldsymbol{\mathcal{H}}_i \mathfrak{q}]_{mn} \\
&= \sum_{\tau,\tau'\in[T]} [\mathfrak{p}]_{\tau m}^{\mathsf{T}} [\boldsymbol{\mathcal{H}}_i]_{\tau\tau'} [\mathfrak{q}]_{\tau' n} \\
&= \sum_{\tau,\tau'\in[T]} \begin{Bmatrix} \mathfrak{p}^{(m)}, & \tau = m; \\ 0_{KM_\tau \times KM_m}, & \text{otherwise} \end{Bmatrix}^{\mathsf{T}} \\
&\qquad \cdot \text{blkdiag}\{\boldsymbol{\mathcal{H}}_{\tau\tau',i}^{(t)}\}_{t\in[T]} \\
&\qquad \cdot \begin{Bmatrix} \mathfrak{q}^{(n)}, & \tau' = n; \\ 0_{KM_{\tau'} \times KM_n}, & \text{otherwise} \end{Bmatrix} \\
&= \mathfrak{p}^{(m)\mathsf{T}} \cdot \text{blkdiag}\{\boldsymbol{\mathcal{H}}_{mn,i}^{(t)}\}_{t\in[T]} \cdot \mathfrak{q}^{(n)} \\
&= (p^{(m)} \otimes I_{M_m})^{\mathsf{T}} \cdot \boldsymbol{\mathcal{H}}_{mn,i}^{(m)} \cdot (\mathbf{1}_{K_m} \otimes I_{M_n}) \\
&= \sum_{k\in\mathcal{N}^{(m)}} p_k^{(m)} \boldsymbol{H}_{k,mn,i}.
\end{aligned}$$

Note that $\boldsymbol{H}_{k,mn,i}$ is of size $M_m \times M_n$. Third, by equating Eqn. (39) with $\mathcal{V}^{\mathsf{H}}\times$ Eqn. (20), we have

$$\begin{aligned}
\mathcal{G}^\star &= \overline{\mathcal{M}}^{-1} \mathcal{V}^{\mathsf{H}} \mathcal{P} \mathcal{M} G^\star \\
&= (\overline{\mathcal{M}}^{-1} \mathcal{J} \overline{\mathcal{M}}) \mathcal{V}^{\mathsf{H}} G^\star = \mathcal{J} \mathcal{V}^{\mathsf{H}} G^\star.
\end{aligned}$$

The last equality is due to $\overline{\mathcal{M}}$ and $\mathcal{J}$ both being block diagonal matrices with identical partition. We have for $m \in [T]$,

$$\begin{aligned}
[\mathcal{G}^\star]_m &= [\mathcal{V}^{\mathsf{H}} G^\star]_m \\
&= \sum_{\tau\in[T]} [\mathcal{V}]_{\tau m}^{\mathsf{H}} [G^\star]_\tau \\
&= \mathfrak{p}^{(m)\mathsf{T}} \text{col}\left\{\text{col}\left\{\nabla_m J_{\ell_k}^{(m)}(x^\star)\right\}_{k\in\mathcal{N}^{(t}}\right\}_{t\in[T]} \\
&= (p^{(m)} \otimes I_{M_m})^{\mathsf{T}} \text{col}\left\{\nabla_m J_{\ell_k}^{(m)}(x^\star)\right\}_{k\in\mathcal{N}^{(m)}} \\
&= \sum_{k\in\mathcal{N}^{(m)}} p_k^{(m)} \nabla_m J_k^{(m)}(x^\star) = 0_{M_m\times 1}.
\end{aligned}$$

Thus, we see that the upper block of $\mathcal{G}^\star$ is identically zero. Last, we have

$$\begin{aligned}
\boldsymbol{\mathcal{S}}_i &= \overline{\mathcal{M}}^{-1} \mathcal{V}^{\mathsf{H}} \mathcal{P} \mathcal{M} \boldsymbol{S}_i \\
&= (\overline{\mathcal{M}}^{-1} \mathcal{J} \overline{\mathcal{M}}) \mathcal{V}^{\mathsf{H}} \boldsymbol{S}_i = \mathcal{J} \mathcal{V}^{\mathsf{H}} \boldsymbol{S}_i
\end{aligned}$$

by identical procedures as before. □

Lemma 6. *The blocks $\boldsymbol{\mathcal{D}}_{12,i}$, $\boldsymbol{\mathcal{D}}_{21,i}$, and $\boldsymbol{\mathcal{D}}_{22,i}$ have finite bounds on their spectral norms independent of $\mu_{\max}$ and i, which are denoted by σ_{12}, σ_{21}, and σ_{22}, respectively. Similarly, $\|\mathcal{G}_2^\star\|$ has a finite bound on its spectral norm independent of $\mu_{\max}$.*

Proof. The independence of the bounds can be seen from the definitions (in the proof of Theorem 2), which do not depend on $\overline{\mathcal{M}}$. The existence of these finite bounds is attributed to the boundedness of $\boldsymbol{\mathcal{H}}_i$. The details are irrelevant to our discussion and are hence omitted. □

The recursion relation in Theorem 2 is the equation that we work with to bound the error moments in the proofs that follow. To bound the terms in Eqn. (39), we require the following lemmas.

Lemma 7 (Block-1-norm and block-∞-norm bound on spectral norm)**.** *Consider any block partition of a matrix A, with the block-1-norm and block-∞-norm defined as*

$$\|A\|_{\text{b},1} := \max_j \sum_i \left\|[A]_{ij}\right\| \tag{41}$$

and

$$\|A\|_{\text{b},\infty} := \max_i \sum_j \left\|[A]_{ij}\right\|. \tag{42}$$

Then, we have the following bound on the spectral norm of A as

$$\|A\| \le \sqrt{\|A\|_{\text{b},1} \cdot \|A\|_{\text{b},\infty}}. \tag{43}$$

Proof. Note that $\|\cdot\|_{\text{b},1}$ is sub-multiplicative:

$$\begin{aligned}
\|AB\|_{\text{b},1} &= \max_j \sum_i \left\|\sum_k [A]_{ik}[B]_{kj}\right\| \\
&\le \max_j \sum_{i,k} \left\|[A]_{ik}[B]_{kj}\right\| \\
&\le \max_j \sum_{i,k} \left\|[A]_{ik}\right\| \left\|[B]_{kj}\right\| \\
&\le \|A\|_{\text{b},1} \max_j \sum_k \left\|[B]_{kj}\right\| = \|A\|_{\text{b},1} \|B\|_{\text{b},1}.
\end{aligned}$$

Therefore, consider an eigenvalue-eigenvector pair (λ, x) of A. We have

$$\begin{aligned}&\because \|A\|_{\mathrm{b},1}\|x\|_{\mathrm{b},1} \geq \|Ax\|_{\mathrm{b},1} = |\lambda|\|x\|_{\mathrm{b},1}\\ &\therefore \|A\|_{\mathrm{b},1} \geq \max|\lambda| = \rho(A),\end{aligned}$$

where ρ is the spectral radius. Finally,

$$\begin{aligned}&\rho(AA^{\mathsf{H}}) \leq \left\|AA^{\mathsf{H}}\right\|_{\mathrm{b},1} \leq \|A\|_{\mathrm{b},1}\left\|A^{\mathsf{H}}\right\|_{\mathrm{b},1} = \|A\|_{\mathrm{b},1}\|A\|_{\mathrm{b},\infty}\\ &\Rightarrow \|A\| = \rho(AA^{\mathsf{H}}) \leq \sqrt{\|A\|_{\mathrm{b},1}\cdot\|A\|_{\mathrm{b},\infty}}.\end{aligned}$$

Thus, the statement is proven. □

Lemma 8 (Spectral norm of $\mathcal{J}_{11} - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i}$)**.** *Under Assumption 4, the spectral norm of the upper-left block of $\mathcal{J} - \overline{\mathcal{M}}\boldsymbol{\mathcal{D}}_i$ can be bounded by*

$$\left\|\mathcal{J}_{11} - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i}\right\| = 1 - \Omega(\mu_{\max}). \tag{44}$$

Proof. Remember that $\mathcal{J}_{11} = I_M$. Let us bound its spectral norm via its block-1-norm and block-∞-norm (Lemma 7):

$$\begin{aligned}&\left\|I_M - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1}\right\|^2\\ &\leq \left\|I_M - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i}\right\|_{\mathrm{b},1}\left\|I_M - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1}\right\|_{\mathrm{b},\infty},\end{aligned}$$

where the partition of the matrices is a $T \times T$ block partition, with the (tt')-block be of size $M_t \times M_{t'}$ for $t, t' \in [T]$. Since

$$[\boldsymbol{\mathcal{D}}_{11,i}]_{tt'} = \sum_{k\in\mathcal{N}^{(t)}} p_k^{(t)}\boldsymbol{H}_{k,tt',i},$$

the diagonal blocks satisfy $[\boldsymbol{\mathcal{D}}_{11,i}]_{tt} \succeq \varepsilon^{(t)} I_{M_t}$ by the fourth point of Assumption 4, and the off-diagonal blocks satisfy $\|[\boldsymbol{\mathcal{D}}_{11,i}]_{tt'}\| \leq \delta^{(t't)}$ by the fifth point of Assumption 4.

Finally, we arrive at

$$\begin{aligned}&\left\|I_M - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1}\right\|^2\\ &\leq \max_{t'\in[T]}\left(\left\|I_M - \mu^{(t')}[\boldsymbol{\mathcal{D}}_{11,i}]_{t't'}\right\| + \sum_{t\neq t'}\left\|\mu^{(t)}[\boldsymbol{\mathcal{D}}_{11,i}]_{tt'}\right\|\right)\\ &\quad\cdot \max_{t\in[T]}\left(\left\|I_M - \mu^{(t)}[\boldsymbol{\mathcal{D}}_{11,i}]_{tt}\right\| + \sum_{t'\neq t}\left\|\mu^{(t)}[\boldsymbol{\mathcal{D}}_{11,i}]_{tt'}\right\|\right)\\ &\leq \max_{t'\in[T]}\left(1 - \mu^{(t')}\varepsilon^{(t')} + \sum_{t\neq t'}\mu^{(t)}\delta^{(tt')}\right)\\ &\quad\cdot \max_{t\in[T]}\left(1 - \mu^{(t)}\varepsilon^{(t)} + \mu^{(t)}\sum_{t'\neq t}\delta^{(tt')}\right)\\ &= 1 - \Theta(\mu_{\max}).\end{aligned}$$

Thus it is shown. Note that the second inequality relies on the sixth point of Assumption 4, and it holds only for $\mu_{\max}$ *small enough*. The exact characterization of how small $\mu_{\max}$ should be depends on the spectrum of $[\boldsymbol{\mathcal{D}}_{11,i}]_{tt}$. □

Lemma 9. *For $t_k \in (0,1)$ such that $t := \sum_{k=1}^N t_k < 1$,*

$$\left\|\sum_{k=0}^N a_n\right\|^2 \leq \frac{1}{1-t}\|a_0\|^2 + \sum_{k=1}^N \frac{1}{t_k}\|a_k\|^2. \tag{45}$$

Proof. Since $\|\cdot\|^2$ is a convex function, apply Jensen's inequality to

$$\left\|\sum_{k=0}^N a_n\right\|^2 = \left\|(1-t)\cdot\frac{a_0}{1-t} + \sum_{k=1}^N t_k\cdot\frac{a_k}{t_k}\right\|^2.$$

The lemma is then proven. □

Lemma 10 (Bounded Second Moment of Gradient Noise)**.** *Under Assumption 5, the following holds for all $t \in [T]$ and all $k \in \mathcal{N}$:*

$$\mathbb{E}\left[\left\|\boldsymbol{s}_{k,i}^{(t)}\right\|^2 \middle| \mathcal{F}_{i-1}\right] \leq \left(\overline{\beta}_k^{(t)}\right)^2\left\|\boldsymbol{x}_{k,i-1}\right\|^2 + \left(\overline{\sigma}_k^{(t)}\right)^2. \tag{46}$$

Proof. By Jensen's inequality, $(\mathbb{E}[\|s\|^2])^2 \leq \mathbb{E}[\|s\|^4]$. Further note that for any positive a and b, we have $a^2 + b^2 \leq (a+b)^2$. □

Lemma 11 (Bounded Moments Centered about Nash Equilibrium)**.** *Under Assumption 5, the following equations also hold:*

$$\mathbb{E}\left[\left\|\boldsymbol{s}_{k,i}^{(t)}\right\|^4 \middle| \mathcal{F}_{i-1}\right] \leq \left(\beta_{4,k}^{(t)}\right)^4\left\|\widetilde{\boldsymbol{x}}_{k,i-1}\right\|^4 + \left(\sigma_{4,k}^{(t)}\right)^4, \tag{47a}$$

$$\mathbb{E}\left[\left\|\boldsymbol{s}_{k,i}^{(t)}\right\|^2 \middle| \mathcal{F}_{i-1}\right] \leq \left(\beta_{2,k}^{(t)}\right)^2\left\|\widetilde{\boldsymbol{x}}_{k,i-1}\right\|^2 + \left(\sigma_{2,k}^{(t)}\right)^2 \tag{47b}$$

for some constants $\beta_{4,k}^{(t)}$, $\sigma_{4,k}^{(t)}$, $\beta_{2,k}^{(t)}$, and $\sigma_{2,k}^{(t)}$.

Proof. We only proof the first equation, the second equation can be proven by similar procedures. Since for any vectors a and b, we can apply Jensen's inequality on $\|\cdot\|^4$ to obtain

$$\|a+b\|^4 = \left\|\frac{1}{2}\cdot 2a + \frac{1}{2}\cdot 2b\right\|^4 \leq 8\|a\|^4 + 8\|b\|^4,$$

we have

$$\begin{aligned}&\left(\overline{\beta}_k^{(t)}\right)^4\left\|\boldsymbol{x}_{k,i-1}\right\|^4 + \left(\overline{\sigma}_k^{(t)}\right)^4\\ &= \left(\overline{\beta}_k^{(t)}\right)^4\left\|\boldsymbol{x}_{k,i-1} - x^\star - x^\star\right\|^4 + \left(\overline{\sigma}_k^{(t)}\right)^4\\ &\leq \underbrace{8\left(\overline{\beta}_k^{(t)}\right)^4}_{=:\left(\beta_{4,k}^{(t)}\right)^4}\left\|\widetilde{\boldsymbol{x}}_{k,i-1}\right\|^4 + \underbrace{8\left(\overline{\beta}_k^{(t)}\right)^4\|x^\star\|^4 + \left(\overline{\sigma}_k^{(t)}\right)^4}_{=:\left(\sigma_{4,k}^{(t)}\right)^4}.\end{aligned}$$

Thus it is proven. □

C. SECOND MOMENT STABILITY

Our goal is to analyze the behavior and upper-bound the growth to $\|\widetilde{\boldsymbol{X}}_i\|^2$. But since

$$\left\|\widetilde{\boldsymbol{X}}_i\right\|^2 = \left\|\mathcal{V}^{-\mathsf{H}}\widetilde{\boldsymbol{W}}_i\right\|^2 \leq \left\|\mathcal{V}^{-1}\right\|^2 \left\|\widetilde{\boldsymbol{W}}_i\right\|^2 = \left\|\mathcal{V}^{-1}\right\|^2 \left(\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^2 + \left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^2\right),$$

we can change our goal to upper-bound the growth to $\widetilde{\boldsymbol{W}}_{1,i}$ and $\widetilde{\boldsymbol{W}}_{2,i}$ instead. Let us start the proof from the error recursion in the Perron coordinates as specified in Theorem 2.

First, let us analyze $\|\widetilde{\boldsymbol{W}}_{1,i}\|^2$: taking the expectation conditioned on the filtration $\mathcal{F}_{i-1}$ first, we have

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^2 \middle| \mathcal{F}_{i-1}\right] \\
&= \left\|(\mathcal{J}_{11} - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1})\widetilde{\boldsymbol{W}}_{1,i-1} - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{12,i-1}\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2 \\
&\quad + \mathbb{E}\left[\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{S}}_{1,i}\right\|^2 \middle| \mathcal{F}_{i-1}\right].
\end{aligned}$$

Then, invoke Assumption 5 and take the expectation over the filtration, we have

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^2\right] \\
&= \mathbb{E}\left[\left\|(\mathcal{J}_{11} - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1})\widetilde{\boldsymbol{W}}_{1,i-1} \right.\right. \\
&\qquad \left.\left. -\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{12,i-1}\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right] \\
&\quad + \mathbb{E}\left[\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{S}}_{1,i}\right\|^2\right].
\end{aligned}$$

Next, apply Lemma 9 with $t = \alpha\mu_{\max}$, where α is subject to $\left\|\mathcal{J}_{11} - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i}\right\| \leq 1 - \alpha\mu_{\max}$ for all i and whose existence is guaranteed by Lemma 8, and incorporate Lemma 6 to obtain

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^2\right] \\
&\leq \mathbb{E}\left[\frac{\left\|\mathcal{J}_{11} - \mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1}\right\|^2}{1-t}\left\|\widetilde{\boldsymbol{W}}_{1,i-1}^{(1)}\right\|^2\right] \\
&\quad + \mathbb{E}\left[\frac{\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{12,i-1}\right\|^2}{t}\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right] \\
&\quad + \|\mathcal{M}_{11}\|^2\,\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^2\right] \\
&\leq (1-\alpha\mu_{\max})\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i-1}^{(1)}\right\|^2\right] \\
&\quad + \frac{\mu_{\max}\sigma_{12}^2}{\alpha}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right] \\
&\quad + \mu_{\max}^2\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^2\right].
\end{aligned}$$

Next, let us analyze $\|\widetilde{\boldsymbol{W}}_{2,i}\|^2$: taking the expectation conditioned on the filtration $\mathcal{F}_{i-1}$, then take the expectation over it, we have

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^2\right] \\
&= \mathbb{E}\left[\left\|-\mathcal{M}_{22}\boldsymbol{\mathcal{D}}_{21,i-1}\widetilde{\boldsymbol{W}}_{1,i-1} \right.\right. \\
&\qquad \left.\left. +(\mathcal{J}_{22} - \mathcal{M}_{22}\boldsymbol{\mathcal{D}}_{22,i-1})\widetilde{\boldsymbol{W}}_{2,i-1} + \mathcal{M}_{22}\mathcal{G}_2^\star\right\|^2\right] \\
&\quad + \mathbb{E}\left[\left\|\mathcal{M}_{22}\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right].
\end{aligned}$$

To apply Lemma 9 to bound the right-hand side, we need to first bound $\|\mathcal{J}_{22}\|$. Since $\mathcal{J}_{22}$ is a block diagonal matrix, with each diagonal block being $J_{\varepsilon_\tau} \otimes I_{M_\tau}$ ($\tau \in [T]$), we have that

$$\|\mathcal{J}_{22}\|^2 = \max_{\tau\in[T]}\|J_{\varepsilon_\tau}\|^2 = \max_{\tau\in[T]}\lambda_{\max}\left(J_{\varepsilon_\tau}^{\mathsf{H}}J_{\varepsilon_\tau}\right).$$

Note that J_{ε_τ} has ε_τ as its first upper-diagonal, we have the tri-diagonal matrix

$$J_{\varepsilon_\tau}^{\mathsf{H}}J_{\varepsilon_\tau} = \begin{bmatrix} \left|\lambda_{2,\tau}\right|^2 & \varepsilon_\tau\lambda_{2,\tau}^* & \\ \varepsilon_\tau\lambda_{2,\tau} & \left|\lambda_{2,\tau}\right|^2 + \varepsilon_\tau^2 & \ddots \\ & \ddots & \ddots \end{bmatrix}.$$

We can upper-bound its largest eigenvalue (its spectral radius) by any matrix norm, we hence use the 1-norm to upper-bound it and obtain

$$\|\mathcal{J}_{22}\|^2 \leq \max_{\tau\in[T]}\left\|J_{\varepsilon_\tau}^{\mathsf{H}}J_{\varepsilon_\tau}\right\|_1 \leq \max_{\tau\in[T]}\left(\left|\lambda_{2,\tau}\right| + \varepsilon_\tau\right)^2.$$

We denote the maximum of the equation above as $\|\mathcal{J}_{22}\| \leq 1-\gamma$, with $\gamma \in (0,1)$ being a parameter depending solely on $\mathcal{P}$ and the ε_τ's chosen. This also gives an upper bound on the ε_τ's. Next, apply Lemma 9 with $t = 1-\gamma$ and incoporate Lemma 6 to obtain

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^2\right] \\
&\leq \frac{\|\mathcal{J}_{22}\|^2}{t}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right] \\
&\quad + \frac{3\|\mathcal{M}_{22}\|^2}{1-t}\left(\mathbb{E}\left[\left\|\boldsymbol{\mathcal{D}}_{21,i-1}\right\|^2\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^2\right]\right. \\
&\qquad \left. +\mathbb{E}\left[\left\|\boldsymbol{\mathcal{D}}_{22,i-1}\right\|^2\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right] + \|\mathcal{G}_2^\star\|^2\right) \\
&\quad + \|\mathcal{M}_{22}\|^2\,\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right] \\
&\leq (1-\gamma)\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]
\end{aligned}$$

$$+\frac{3\mu_{\max}^2}{\gamma}\left(\sigma_{21}^2\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^2\right]\right.$$
$$\left.+\sigma_{22}^2\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]+\|\mathcal{G}_2^\star\|^2\right)$$
$$+\mu_{\max}^2\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right].$$

The last two terms to be analyzed are the gradient noises. Note that by Assumption 5 and the derived Lemma 11, we have

$$\begin{aligned}
&\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^2\right]+\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right]=\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{i}\right\|^2\right]\\
&\leq\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^2\mathbb{E}\left[\|\boldsymbol{S}_i\|^2\right]\\
&=\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^2\sum_{t\in[T],k\in\mathcal{N}}\mathbb{E}\left[\left\|\boldsymbol{s}_{k,i}^{(t)}\right\|^2\right]\\
&\leq\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^2\sum_{t\in[T],k\in\mathcal{N}}\left(\left(\beta_{2,k}^{(t)}\right)^2\|\widetilde{\boldsymbol{x}}_{k,i-1}\|^2+\left(\sigma_{2,k}^{(t)}\right)^2\right)\\
&\leq\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^2\left(\beta_2^2\left(\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^2+\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right)+\sigma_2^2\right),
\end{aligned}$$

for some constants $\beta_2^2=\left\|\mathcal{V}^{-1}\right\|^2\max_{t,k}(\beta_{2,k}^{(t)})^2$ and $\sigma_2^2=KM\max_{t,k}(\sigma_{2,k}^{(t)})^2$. The gradient noise terms thus contribute $O(\mu_{\max}^2)$ to the recursions.

Combining the two second moment recursions of the expected error in the transformed coordinates above, we have

$$\widetilde{\mathcal{W}}_i^2\leq\Gamma\cdot\widetilde{\mathcal{W}}_{i-1}^2+\begin{bmatrix}O(\mu_{\max}^2)\\O(\mu_{\max}^2)\end{bmatrix}$$

with $\widetilde{\mathcal{W}}_i^2:=\mathrm{col}\{\mathbb{E}[\,\|\widetilde{\boldsymbol{W}}_{1,i}\|^2],\mathbb{E}[\,\|\widetilde{\boldsymbol{W}}_{2,i}\|^2]\}\in\mathbb{R}^{2\times1}$ and

$$\Gamma=\begin{bmatrix}1-O(\mu_{\max})&O(\mu_{\max})\\O(\mu_{\max}^2)&1-\gamma+O(\mu_{\max}^2)\end{bmatrix}.$$

The matrix Γ is stable as we can bound its spectral radius by its 1-norm:

$$\rho(\Gamma)\leq\max\left\{\begin{matrix}1-O(\mu_{\max})+O(\mu_{\max}^2),\\1-\gamma+O(\mu_{\max})+O(\mu_{\max}^2)\end{matrix}\right\},$$

we have $\rho(\Gamma)<1$ for $\mu_{\max}$ small enough. The combined recursion above is element-wise upper bounded by

$$\begin{aligned}
\limsup_{i\to\infty}\widetilde{\mathcal{W}}_i^2&\leq(I-\Gamma)^{-1}\cdot\begin{bmatrix}O(\mu_{\max}^2)\\O(\mu_{\max}^2)\end{bmatrix}\\
&=\begin{bmatrix}O(1/\mu_{\max})&O(1)\\O(\mu_{\max})&O(1)\end{bmatrix}\begin{bmatrix}O(\mu_{\max}^2)\\O(\mu_{\max}^2)\end{bmatrix}\\
&=\begin{bmatrix}O(\mu_{\max})\\O(\mu_{\max}^2)\end{bmatrix}.
\end{aligned}$$

We thus obtained the second moment stability:

$$\begin{aligned}
&\limsup_{i\to\infty}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_i\right\|^2\right]=O(\mu_{\max})+O(\mu_{\max}^2)=O(\mu_{\max})\\
\Rightarrow&\limsup_{i\to\infty}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{X}}_i\right\|^2\right]=O(\mu_{\max}).
\end{aligned}$$

D. FOURTH MOMENT STABILITY

The proof for the stability of the fourth moment error recursion introduces nothing new, as it follows directly like that in [1]. One simply goes back to Eqn. (39) and directly compute the expectation to the fourth moment of $\widetilde{\boldsymbol{W}}_{1,i}$ and $\widetilde{\boldsymbol{W}}_{2,i}$. The bound to each term is mostly exactly the same as that in the previous section. For the sake of completeness, we gloss over the key steps of the proof.

The first step is to utilize the following inequality

$$\|a+b\|^4\leq\|a\|^4+3\|b\|^4+8\|a\|^2\|b\|^2+4\|a\|^2\,\mathbf{Re}\{a^*b\}$$

to obtain a bound on the fourth moments. The second step is to take the expectation conditioned on the filtration $\mathcal{F}_{i-1}$ then take the expectation over it, obtaining

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^4\right]\\
&\leq\mathbb{E}\left[\left\|(\mathcal{J}_{11}-\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1})\widetilde{\boldsymbol{W}}_{1,i-1}\right.\right.\\
&\qquad\left.\left.-\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{12,i-1}\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right]\\
&\quad+3\mathbb{E}\left[\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{S}}_{1,i}\right\|^4\right]\\
&\quad+8\mathbb{E}\left[\left\|(\mathcal{J}_{11}-\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1})\widetilde{\boldsymbol{W}}_{1,i-1}\right.\right.\\
&\qquad\left.\left.-\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{12,i-1}\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]\mathbb{E}\left[\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{S}}_{1,i}\right\|^2\right].
\end{aligned}$$

Next, adapt Lemma 9 to $\|\cdot\|^4$ and set $t=\alpha\mu_{\max}$ to obtain

$$\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^4\right]\\
&\leq\mathbb{E}\left[\frac{\left\|\mathcal{J}_{11}-\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1}\right\|^4}{(1-t)^3}\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^4\right]\\
&\quad+\mathbb{E}\left[\frac{\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{12,i-1}\right\|^4}{t^3}\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right]\\
&\quad+3\mathbb{E}\left[\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{S}}_{1,i}\right\|^4\right]\\
&\quad+8\left(\mathbb{E}\left[\frac{\left\|\mathcal{J}_{11}-\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{11,i-1}\right\|^2}{1-t}\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^2\right]\right.
\end{aligned}$$

$$
\begin{aligned}
&+\mathbb{E}\left[\frac{\left\|\mathcal{M}_{11}\boldsymbol{\mathcal{D}}_{12,i-1}\right\|^2}{t}\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]\Bigg)\\
&\cdot\|\mathcal{M}_{11}\|^2\,\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^2\right]\\
\le\ &(1-\alpha\mu_{\max})\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^4\right]\\
&+\frac{\mu_{\max}\sigma_{12}^4}{\alpha^3}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right]+3\mu_{\max}^4\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^4\right]\\
&+8\mu_{\max}^2\Bigg((1-\alpha\mu_{\max})\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^2\right]\\
&\quad+\frac{\mu_{\max}\sigma_{12}^2}{\alpha}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]\Bigg)\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^2\right].
\end{aligned}
$$

Similarly, we have

$$
\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^4\right]\\
&\le\mathbb{E}\Big[\Big\|-\mathcal{M}_{22}\boldsymbol{\mathcal{D}}_{21,i-1}\widetilde{\boldsymbol{W}}_{1,i-1}\\
&\qquad+(\mathcal{J}_{22}-\mathcal{M}_{22}\boldsymbol{\mathcal{D}}_{22,i-1})\widetilde{\boldsymbol{W}}_{2,i-1}+\mathcal{M}_{22}\mathcal{G}_2^\star\Big\|^4\Big]\\
&\quad+3\mathbb{E}\left[\left\|\mathcal{M}_{22}\boldsymbol{\mathcal{S}}_{2,i}\right\|^4\right]\\
&\quad+8\mathbb{E}\left[\left\|\mathcal{M}_{22}\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right]\mathbb{E}\Big[\Big\|-\mathcal{M}_{22}\boldsymbol{\mathcal{D}}_{21,i-1}\widetilde{\boldsymbol{W}}_{1,i-1}\\
&\qquad+(\mathcal{J}_{22}-\mathcal{M}_{22}\boldsymbol{\mathcal{D}}_{22,i-1})\widetilde{\boldsymbol{W}}_{2,i-1}+\mathcal{M}_{22}\mathcal{G}_2^\star\Big\|^2\Big].
\end{aligned}
$$

Apply the adapted Lemma 9 and set $t=1-\gamma$ to obtain

$$
\begin{aligned}
&\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^4\right]\\
&\le\frac{\|\mathcal{J}_{22}\|^4}{t^3}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right]\\
&\quad+\frac{27\|\mathcal{M}_{22}\|^4}{(1-t)^3}\Bigg(\mathbb{E}\left[\left\|\boldsymbol{\mathcal{D}}_{21,i-1}\right\|^4\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^4\right]\\
&\qquad+\mathbb{E}\left[\left\|\boldsymbol{\mathcal{D}}_{22,i-1}\right\|^4\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right]+\|\mathcal{G}_2^\star\|^4\Bigg)\\
&\quad+3\|\mathcal{M}_{22}\|^4\,\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^4\right]\\
&\quad+8\|\mathcal{M}_{22}\|^2\,\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right]\Bigg(\frac{\|\mathcal{J}_{22}\|^2}{t}\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]\\
&\qquad+\frac{3\|\mathcal{M}_{22}\|^2}{1-t}\Bigg(\mathbb{E}\left[\left\|\boldsymbol{\mathcal{D}}_{21,i-1}\right\|^2\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^2\right]\\
&\qquad\quad+\mathbb{E}\left[\left\|\boldsymbol{\mathcal{D}}_{22,i-1}\right\|^2\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]+\|\mathcal{G}_2^\star\|^2\Bigg)\Bigg)\\
&\le(1-\gamma)\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right]\\
&\quad+\frac{27\mu_{\max}^4}{\gamma^3}\Bigg(\sigma_{21}^4\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^4\right]\\
&\qquad+\sigma_{22}^4\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right]+\|\mathcal{G}_2^\star\|^4\Bigg)\\
&\quad+3\mu_{\max}^4\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^4\right]\\
&\quad+8\mu_{\max}^2\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right]\Bigg((1-\gamma)\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]\\
&\qquad+\frac{3\mu_{\max}^2}{\gamma}\Bigg(\sigma_{21}^2\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^2\right]\\
&\qquad\quad+\sigma_{22}^2\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^2\right]+\|\mathcal{G}_2^\star\|^2\Bigg)\Bigg).
\end{aligned}
$$

The third step is to bound the fourth order gradient noise. By Assumption 5 and the derived Lemma 11, we have

$$
\begin{aligned}
&\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^4\right]+\mathbb{E}\left[\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^4\right]\\
&\le\mathbb{E}\left[\left(\left\|\boldsymbol{\mathcal{S}}_{1,i}\right\|^2+\left\|\boldsymbol{\mathcal{S}}_{2,i}\right\|^2\right)^2\right]=\mathbb{E}\left[\|\boldsymbol{\mathcal{S}}_i\|^4\right]\\
&\le\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\mathbb{E}\left[\|\boldsymbol{S}_i\|^4\right]\\
&=\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\mathbb{E}\left[\left(\sum_{t\in[T],k\in\mathcal{N}}\left\|\boldsymbol{s}_{k,i-1}^{(t)}\right\|^2\right)^2\right]\\
&\le KM\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\sum_{t\in[T],k\in\mathcal{N}}\mathbb{E}\left[\left\|\boldsymbol{s}_{k,i-1}^{(t)}\right\|^4\right]\\
&\le KM\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\sum_{t,k}\left(\left(\beta_{4,k}^{(t)}\right)^4\|\widetilde{\boldsymbol{x}}_{k,i-1}\|^4+\left(\sigma_{4,k}^{(t)}\right)^4\right)\\
&\le KM\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\Bigg(\max_{t,k}\left(\beta_{4,k}^{(t)}\right)^4\sum_{t,k}\|\widetilde{\boldsymbol{x}}_{k,i-1}\|^4\\
&\qquad+KM\max_{t,k}\left(\sigma_{4,k}^{(t)}\right)^4\Bigg)\\
&\le KM\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\left(\max_{t,k}\left(\beta_{4,k}^{(t)}\right)^4\left\|\widetilde{\boldsymbol{X}}_{i-1}\right\|^4+\sigma_4^4\right)\\
&\le KM\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\left(\frac{1}{2}\beta_4^4\left\|\widetilde{\boldsymbol{W}}_{i-1}\right\|^4+\sigma_4^4\right)\\
&\le KM\left\|\mathcal{J}\mathcal{V}^{\mathsf{H}}\right\|^4\left(\beta_4^4\left(\left\|\widetilde{\boldsymbol{W}}_{1,i-1}\right\|^4+\left\|\widetilde{\boldsymbol{W}}_{2,i-1}\right\|^4\right)+\sigma_4^4\right)
\end{aligned}
$$

for some constants $\beta_4^4=2\left\|\mathcal{V}^{-1}\right\|^4\max_{t,k}(\beta_{4,k}^{(t)})^4$ and $\sigma_4^4=KM\max_{t,k}(\sigma_{4,k}^{(4)})^4$.

Before we combine the two fourth moment recursions into a single matrix recursion relation, we need to deal with the

product terms between $\widetilde{\boldsymbol{W}}_{i-1}$ and $\boldsymbol{\mathcal{S}}_i$. For any real vectors a and b, we have

$$2\|a\|^2\|b\|^2 \leq \|a\|^4 + \|b\|^4 .$$

Henceforth, we obtain

$$\left(\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^2\right]\right)^2 \leq \mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^4\right],$$
$$\left(\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^2\right]\right)^2 \leq \mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^4\right],$$

and

$$2\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^2\right]\mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^2\right] \leq \mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{1,i}\right\|^4\right] + \mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_{2,i}\right\|^4\right].$$

These relations are plugged back into the two fourth moment recursions. Combining the two fourth moment recursions of the expected error in the transformed coordinates above, we have

$$\widetilde{\mathcal{W}}_i^4 \leq \Gamma \cdot \widetilde{\mathcal{W}}_{i-1}^4 + \Gamma' \cdot \widetilde{\mathcal{W}}_i^2 + \begin{bmatrix} O(\mu_{\max}^4) \\ O(\mu_{\max}^4) \end{bmatrix}$$

with $\widetilde{\mathcal{W}}_i^4 := \mathrm{col}\{\mathbb{E}[\,\|\widetilde{\boldsymbol{W}}_{1,i}\|^4], \mathbb{E}[\,\|\widetilde{\boldsymbol{W}}_{2,i}\|^4]\} \in \mathbb{R}^{2\times 1}$ and

$$\Gamma = \begin{bmatrix} 1 - O(\mu_{\max}) & O(\mu_{\max}) \\ O(\mu_{\max}^2) & 1 - \gamma + O(\mu_{\max}^2) \end{bmatrix},$$
$$\Gamma' = \begin{bmatrix} O(\mu_{\max}^2) & O(\mu_{\max}^3) \\ O(\mu_{\max}^4) & O(\mu_{\max}^2) \end{bmatrix}.$$

The matrix Γ is again stable. Moreover, by the second moment stability, we have

$$\limsup_{i\to\infty} \Gamma' \cdot \widetilde{\mathcal{W}}_i^2 = \begin{bmatrix} O(\mu_{\max}^2) & O(\mu_{\max}^3) \\ O(\mu_{\max}^4) & O(\mu_{\max}^2) \end{bmatrix} \begin{bmatrix} O(\mu_{\max}) \\ O(\mu_{\max}^2) \end{bmatrix} = \begin{bmatrix} O(\mu_{\max}^3) \\ O(\mu_{\max}^4) \end{bmatrix}.$$

Therefore, we conclude that

$$\limsup_{i\to\infty} \widetilde{\mathcal{W}}_i^4 \leq (I-\Gamma)^{-1} \cdot \begin{bmatrix} O(\mu_{\max}^3) \\ O(\mu_{\max}^4) \end{bmatrix} = \begin{bmatrix} O(1/\mu_{\max}) & O(1) \\ O(\mu_{\max}) & O(1) \end{bmatrix} \begin{bmatrix} O(\mu_{\max}^3) \\ O(\mu_{\max}^4) \end{bmatrix} = \begin{bmatrix} O(\mu_{\max}^2) \\ O(\mu_{\max}^4) \end{bmatrix}.$$

We thus obtained the fourth moment stability:

$$\limsup_{i\to\infty} \mathbb{E}\left[\left\|\widetilde{\boldsymbol{W}}_i\right\|^4\right] = O(\mu_{\max}^2) + O(\mu_{\max}^4) = O(\mu_{\max}^2)$$
$$\Rightarrow \limsup_{i\to\infty} \mathbb{E}\left[\left\|\widetilde{\boldsymbol{X}}_i\right\|^4\right] = O(\mu_{\max}^2).$$

E. ANCILLARY LEMMAS FOR FIRST MOMENT STABILITY

The proof of the first moment stability requires additional lemmas and theorems, which we introduce in this section.

Denote the Hessians evaluated at the Nash equilibrium to be $\mathcal{H}$, which is defined as

$$[\mathcal{H}]_{\tau\tau' tt'} := \begin{cases} \mathcal{H}_{\tau\tau'}^{(t)}, & t = t'; \\ 0_{K_t M_\tau \times K_{t'} M_{\tau'}}, & \text{otherwise}, \end{cases} \tag{48a}$$
$$\mathcal{H}_{\tau\tau'}^{(t)} := \mathrm{blkdiag}\{H_{k,\tau\tau'}\}_{k\in\mathcal{N}^{(t)}}, \tag{48b}$$
$$H_{k,\tau\tau'} := \nabla_{\tau'\tau}^2 J_{\ell_k}^{(\tau)}(x^\star). \tag{48c}$$

We define the Hessian deviation to be

$$\widetilde{\boldsymbol{\mathcal{H}}}_{i-1} := \mathcal{H} - \boldsymbol{\mathcal{H}}_{i-1}. \tag{49}$$

Then, error recursion of Eqn. (20) implies the *mean-error recursion*:

$$\mathbb{E}\left[\widetilde{\boldsymbol{X}}_i\right] = \mathcal{B} \cdot \mathbb{E}\left[\widetilde{\boldsymbol{X}}_{i-1}\right] + \mathcal{P}\mathcal{M}E_{i-1} + \mathcal{P}\mathcal{M}G^\star, \tag{50}$$

where

$$\mathcal{B} := \mathcal{P}(I - \mathcal{M}\mathcal{H}) \tag{51}$$

and

$$E_{i-1} := \mathbb{E}\left[\widetilde{\boldsymbol{\mathcal{H}}}_{i-1}\widetilde{\boldsymbol{X}}_{i-1}\right]. \tag{52}$$

Then, multiple both sides of Eqn. (50) by $\mathcal{V}^{\mathsf{H}}$ to obtain:

Theorem 3 (Coordinate transform on mean-error recursion)**.** *Consider the coordinate transform $\widetilde{\boldsymbol{W}}_i = \mathcal{V}^{\mathsf{H}}\widetilde{\boldsymbol{X}}_i$. Denote $\widetilde{Z}_i = \mathbb{E}[\widetilde{\boldsymbol{W}}_i] = \mathrm{col}\{\widetilde{Z}_{1,i}, \widetilde{Z}_{2,i}\}$, $\widetilde{Z}_{1,i} = \mathbb{E}[\widetilde{\boldsymbol{W}}_{1,i}]$, and $\widetilde{Z}_{2,i} = \mathbb{E}[\widetilde{\boldsymbol{W}}_{2,i}]$, then*

$$\widetilde{Z}_i = (\mathcal{J} - \overline{\mathcal{M}}\mathcal{D})\widetilde{Z}_{i-1} + \overline{\mathcal{M}}\mathcal{E}_{i-1} + \overline{\mathcal{M}}\mathcal{G}^\star, \tag{53}$$

where $\mathcal{J}$, $\overline{\mathcal{M}}$, and $\mathcal{G}^\star$ are as defined in Theorem 2, and

$$\mathcal{D} := \mathcal{V}^{\mathsf{H}}\mathcal{H}\mathcal{V}^{-\mathsf{H}} = \begin{bmatrix} \mathcal{D}_{11} & \mathcal{D}_{12} \\ \mathcal{D}_{21} & \mathcal{D}_{22} \end{bmatrix}, \tag{54}$$
$$\mathcal{E}_i := \mathcal{J}\mathcal{V}^{\mathsf{H}}E_i = \begin{bmatrix} \mathcal{E}_{1,i} \\ \mathcal{E}_{2,i} \end{bmatrix}. \tag{55}$$

The block partitions of the matrices are consistent with that of Theorem 2.

Proof. We only need to demonstrate that the definitions and block partitions of $\mathcal{D}$ and $\mathcal{E}_i$ are consistent.

Plug in the partition of $\mathcal{V}$ into the definition of $\mathcal{D}$, we have

$$\mathcal{D} := \mathcal{V}^{\mathsf{H}}\mathcal{H}\mathcal{V}^{-\mathsf{H}} = \begin{bmatrix} \mathfrak{p}^{\mathsf{T}} \\ \mathcal{V}_R^{\mathsf{H}} \end{bmatrix} \mathcal{H} \begin{bmatrix} \mathfrak{q} & \mathcal{V}_L \end{bmatrix}.$$

Similar to the details of $\boldsymbol{\mathcal{D}}_i$, the four blocks are analyzed separately. We specifically take a look at the upper-left block

as it has a clean closed form expression. For $m, n \in [T]$, by similar calculations shown in the proof of Theorem 2 for $[\boldsymbol{\mathcal{D}}_{11,i}]_{mn}$, we have

$$[\mathcal{D}_{11}]_{mn} = \sum_{k \in \mathcal{N}^{(m)}} p_k^{(m)} \, \boldsymbol{H}_{k,mn,i}\big|_{\boldsymbol{x}_{k,i}=x^\star} = \nabla_{nm} J^{(m)}(x^\star).$$

Note that at $\boldsymbol{x}_{k,i} = x^\star$, the Hessians are deterministic. The same applies to the other three blocks of $\mathcal{D}$. Therefore, we see that

$$\begin{aligned}\mathcal{D}_{11} &= \begin{bmatrix} \nabla_1^2 J^{(1)}(x^\star) & \nabla_{21}^2 J^{(1)}(x^\star) & \cdots \\ \nabla_{12}^2 J^{(2)}(x^\star) & \nabla_2^2 J^{(2)}(x^\star) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\ &= \nabla_x \begin{bmatrix} \nabla_1 J^{(1)}(x^\star) \\ \nabla_2 J^{(2)}(x^\star) \\ \vdots \\ \nabla_T J^{(T)}(x^\star) \end{bmatrix} = \nabla_x F(x^\star),\end{aligned}$$

where F is the global gradient defined in Eqn. (12).

Finally, we also have

$$\begin{aligned}\mathcal{E}_i &= \overline{\mathcal{M}}^{-1} \mathcal{V}^{\mathsf{H}} \mathcal{P} \mathcal{M} E_i \\ &= (\overline{\mathcal{M}}^{-1} \mathcal{J} \overline{\mathcal{M}}) \mathcal{V}^{\mathsf{H}} E_i = \mathcal{J} \mathcal{V}^{\mathsf{H}} E_i\end{aligned}$$

by identical procedures as before. □

The first step is to analyze the mean-error recursion is to establish the stability of $\mathcal{B}$. Along the way, we establish two key results: the first one is the spectrum of $\mathcal{M}_{11}\mathcal{D}_{11}$; the second one is a strong monotonicity condition on F, which is crucial for the existence and uniqueness of the Nash equilibrium.

Theorem 4 (Spectrum of $\mathcal{M}_{11}\mathcal{D}_{11}$)**.** *Under Assumption 4, the eigenvalues to $\mathcal{M}_{11}\mathcal{D}_{11} = \mathcal{M}_{11}\nabla_x F(x^\star)$ are within the region:*

$$\left\{ \lambda \in \mathbb{C} \,\middle|\, \nu_R \le \mathbf{Re}\{\lambda\} \le \nu_S, \big|\mathbf{Im}\{\lambda\}\big| \le \nu_I \right\}, \tag{56}$$

where ν_R, ν_S, and ν_I are bounded positive constants.

Proof. Remember that the $M \times M$ matrix $\mathcal{M}_{11}\mathcal{D}_{11}$ has a $T \times T$ block partition, with the (tt')-block being of size $M_t \times M_{t'}$ and is of the form

$$[\mathcal{M}_{11}\mathcal{D}_{11}]_{tt'} = \mu^{(t)} \nabla_{t't}^2 J^{(t)}(x^\star).$$

From the fourth and fifth point of Assumption 4, the diagonal blocks are symmetric and satisfy

$$\begin{aligned}[\mathcal{M}_{11}\mathcal{D}_{11}]_{tt} &\succeq \mu^{(t)} \varepsilon^{(t)} I_{M_t}, \\ \big\|[\mathcal{M}_{11}\mathcal{D}_{11}]_{tt}\big\| &\le \mu^{(t)} \delta^{(tt)},\end{aligned}$$

while the off-diagonal blocks satisfy

$$\big\|[\mathcal{M}_{11}\mathcal{D}_{11}]_{tt'}\big\| \le \mu^{(t)} \delta^{(tt')}.$$

From these conditions, we can establish a perturbation bound on the eigenvalues. Let us define the whole matrix as the diagonal blocks D perturbed by E from the off-diagonal blocks. Using the block-Gershgorin's theorem, the eigenvalues to $D + E$ lies in the union of the regions

$$\begin{aligned}G_t &:= \left\{ z \in \mathbb{C} \,\middle|\, \min\big|z - \lambda(D_{tt})\big| \le \sum_{t' \ne t} \|E_{tt'}\| \right\} \\ &\subseteq \left\{ z \in \mathbb{C} \,\middle|\, \min\big|z - \lambda(D_{tt})\big| \le \mu^{(t)} \sum_{t' \ne t} \delta^{(tt')} \right\}.\end{aligned}$$

Since the diagonal blocks D_{tt} are symmetric, it has real eigenvalues within the region $[\mu^{(t)}\varepsilon^{(t)}, \mu^{(t)}\delta^{(tt)}]$. We thus obtain a bound on the region of eigenvalues as

$$\begin{aligned}&\lambda\left(\mathcal{M}_{11}\nabla_x F(x)\right) \\ &\subseteq \bigcup_{t \in [T]} \left\{ z \in \mathbb{C} \,\middle|\, \begin{array}{c} \exists\, s \in [\mu^{(t)}\varepsilon^{(t)}, \mu^{(t)}\delta^{(tt)}], \\ |z - s| \le \mu^{(t)} \sum_{t' \ne t} \delta^{(tt')} \end{array} \right\}.\end{aligned}$$

The three constants are

$$\begin{aligned}\nu_R &= \min_{t \in [T]} \mu^{(t)} \left(\varepsilon^{(t)} - \sum_{t' \ne t} \delta^{(tt')} \right), \\ \nu_S &= \max_{t \in [T]} \mu^{(t)} \sum_{t' \in [T]} \delta^{(tt')}, \\ \nu_I &= \max_{t \in [T]} \mu^{(t)} \sum_{t' \ne t} \delta^{(tt')}.\end{aligned}$$

The positivity of ν_R is guaranteed by the sixth point of Assumption 4. Thus the statement is shown. □

Theorem 5 (Strong monotonicity of $\mathcal{M}_{11}F$)**.** *Under Assumption 4, the scaled global gradient $\mathcal{M}_{11}F$ is ν_R-strongly monotonic, i.e.,*

$$\langle \mathcal{M}_{11}F(x_1) - \mathcal{M}_{11}F(x_2), x_1 - x_2 \rangle \ge \nu_R \|x_1 - x_2\|^2 \tag{57}$$

for any x_1, x_2.

Proof. By the smoothness assumed on the local loss functions (second point of Assumption 4), the strong monotonicity of $\mathcal{M}_{11}F$ is equivalently established if we consider the infinitesimal version, showing the positivity of the quadratic form generated by $\mathcal{M}_{11}\nabla_x F(x)$. The quadratic form generated by $\mathcal{M}_{11}\nabla_x F(x)$ is its symmetrized version

$$S(x) := \frac{1}{2}\left[\mathcal{M}_{11}\nabla_x F(x) + \left(\mathcal{M}_{11}\nabla_x F(x)\right)^{\mathsf{T}} \right].$$

Our goal is to show $S(x) \succ 0$ for all x. The matrix S has the $T \times T$ block partition with the (tt')-block being of size $M_t \times M_{t'}$. By the fourth and fifth points of Assumption 4, the diagonal blocks satisfy

$$[S]_{tt} = \mu^{(t)} \nabla_t^2 J^{(t)}(x) \succeq \mu^{(t)} \varepsilon^{(t)} I_{M_t},$$

and the off-diagonal block at position (tt') is of the form

$$[S]_{tt'} = \frac{1}{2}\left(\nabla_{t't}^2 J^{(t)}(x) + \nabla_{t't}^2 J^{(t')}(x)\right),$$

satisfying

$$\left\|[S]_{tt'}\right\| \leq \frac{1}{2}\left(\mu^{(t)}\delta^{(tt')} + \mu^{(t')}\delta^{(t't)}\right) =: \frac{\Delta_{tt'}}{2}.$$

A lower bound to the minimum eigenvalue of S

$$\nu_R := \lambda_{\min}(S) = \min_{\|v\|=1} v^{\mathsf{T}} S v$$

can be established. Let the vector $v = [v_1; \ldots; v_T] \in \mathbb{R}^M$ be the minimizer of the quadratic form above, where $v_t \in \mathbb{R}^{M_t}$, then

$$\begin{aligned}
\nu_R &= \sum_{t,t' \in [T]} v_t^{\mathsf{T}} [S]_{tt'} v_{t'} \\
&= \sum_{t \in [T]} v_t^{\mathsf{T}} [S]_{tt} v_t + \sum_{t \neq t' \in [T]} v_t^{\mathsf{T}} [S]_{tt'} v_{t'} \\
&\geq \sum_{t \in [T]} \mu^{(t)} \varepsilon^{(t)} \|v_t\|^2 - \sum_{t \neq t' \in [T]} \frac{\Delta_{tt'}}{2} \|v_t\| \|v_{t'}\| \\
&\geq \sum_{t \in [T]} \mu^{(t)} \varepsilon^{(t)} \|v_t\|^2 - \sum_{t \neq t' \in [T]} \frac{\Delta_{tt'}}{4} \left(\|v_t\|^2 + \|v_{t'}\|^2\right) \\
&= \sum_{t \in [T]} \mu^{(t)} \varepsilon^{(t)} \|v_t\|^2 - \sum_{t \neq t' \in [T]} \frac{\Delta_{tt'}}{2} \|v_t\|^2 \\
&\leq \sum_{t \in [T]} \|v_t\|^2 \left(\mu^{(t)} \varepsilon^{(t)} - \sum_{t' \neq t} \frac{\Delta_{tt'}}{2}\right) \\
&\leq \min_{t \in [T]} \left(\mu^{(t)} \varepsilon^{(t)} - \sum_{t' \neq t} \frac{\Delta_{tt'}}{2}\right).
\end{aligned}$$

The term in the bracket is demonstrated to be strictly positive by taking the average of the $\mu^{(t)} \times$ Eqn. (10) and Eqn. (11).

Since $v^{\mathsf{T}} S(x) v \geq \nu_R > 0$, we also have $v^{\mathsf{T}} \nabla_x F(x) v \geq \nu_R > 0$. The strong monotonicity of $\mathcal{M}_{11} F$ is thus shown. □

Since $\mathcal{M}_{11} F$ is strongly monotonic and the Nash equilibrium condition is $F = 0$, the game we consider is guaranteed to have a unique Nash equilibrium [17].

Next, let us show the stability of $\mathcal{B}$.

Theorem 6 (Stability of $\mathcal{B}$)**.** *Under Assumption 4, the matrix $\mathcal{B}$ is stable with spectral radius given by*

$$\rho(\mathcal{B}) = 1 - \Theta(\mu_{\max}). \tag{58}$$

Proof. The proof of this theorem again utilizes the block-Gershgorin's theorem and Theorem 4.

Consider the change of coordinates by

$$\mathcal{T} := \mathrm{blkdiag}\{\mu^{1/K} I_M, \mu^{2/K} I_M, \ldots, \mu^{K/K} I_M\},$$

and $\mathcal{V}^{-\mathsf{H}}$, we obtain

$$\mathcal{T}^{-1}\mathcal{V}^{\mathsf{H}}\mathcal{B}\mathcal{V}^{-\mathsf{H}}\mathcal{T} = \mathcal{T}^{-1}(\mathcal{J} - \overline{\mathcal{M}}\mathcal{D})\mathcal{T}$$

$$= \left[\begin{array}{c|ccc}
I_M - \mathcal{M}_{11}\mathcal{D}_{11} & & O(\mu_{\max}^{(K+1)/K}) & \\
\hline
 & \lambda_2 I_M + O(\mu_{\max}) & & O(\mu_{\max}^{1/K}) \\
O(\mu_{\max}^{1/K}) & & \ddots & \\
 & O(\mu_{\max}^{1/K}) & & \lambda_L I_M + O(\mu_{\max})
\end{array}\right],$$

where the λ_k's in the bottom-right block are the eigenvalues to $\mathcal{J}_{22}$. The eigenvalues to $\lambda_k I_M + O(\mu_{\max})$ is perturbed from λ_k by a perturbation of $O(\mu_{\max})$.

Next, by the block-Gershgorin's theorem, we have that the eigenvalues to $\mathcal{B}$ lie in the regions below

$$\left\{ z \in \mathbb{C} \,\middle|\, \min_k \left|z - \lambda_k \left(I_M - \mathcal{M}_{11}\mathcal{D}_{11}\right)\right| \leq O(\mu_{\max}^{(K+1)/K}) \right\}$$
$$\cup \left\{ z \in \mathbb{C} \,\middle|\, \min_{k \in [2:L]} |z - \lambda_k| \leq O(\mu_{\max}^{1/K}) \right\}$$

If each step-size is made small enough, the two sets above are in fact *disjoint*. Hence, the eigenvalue with the largest norm must lie in the first set. Thus, we obtain the spectral radius of $\mathcal{B}$:

$$\rho(\mathcal{B}) \geq 1 - \underbrace{\max_{t \in [T]} \mu^{(t)} \sum_{t' \in [T]} \delta^{(tt')}}_{=\nu_S} - O(\mu_{\max}^{(K+1)/K})$$

$$\rho(\mathcal{B}) \leq 1 - \underbrace{\min_{t \in [T]} \mu^{(t)} \left(\varepsilon^{(t)} - \sum_{t' \neq t} \delta^{(tt')}\right)}_{=\nu_R} + O(\mu_{\max}^{(K+1)/K}).$$

We thus see that $\rho(\mathcal{B})$ is of order $1 - \Theta(\mu_{\max})$. □

F. FIRST MOMENT STABILITY

Let us begin the proof to the first moment stability. The first step is to bound the size of the Hessian deviation $\widetilde{\boldsymbol{\mathcal{H}}}_i$. By invoking the third point of Assumption 4, we have

$$\begin{aligned}
&\left\|\widetilde{\boldsymbol{H}}_{k,\tau\tau',i}\right\| \\
&= \left\|H_{k,\tau\tau'} - \boldsymbol{H}_{k,\tau\tau',i}\right\| \\
&\leq \int_0^1 \left\|\nabla_{\tau'\tau}^2 J_{\ell_k}^{(\tau)}(x^\star) - \nabla_{\tau'\tau}^2 J_{\ell_k}^{(\tau)}(x^\star - \eta \widetilde{\boldsymbol{x}}_{k,i})\right\| \, \mathrm{d}\eta
\end{aligned}$$

$$\leq \int_0^1 \kappa \left\| \eta \widetilde{\boldsymbol{x}}_{k,i} \right\| \mathrm{d}\eta \leq \frac{\kappa}{2} \left\| \widetilde{\boldsymbol{x}}_{k,i} \right\| \leq \frac{\kappa}{2} \left\| \widetilde{\boldsymbol{X}}_i \right\|.$$

For the whole Hessian deviation, we bound its spectral norm by its block-1-norm and block-∞-norm, where we consider the partition to the level of each $(\tau\tau', tt')$-block being of size $M_t \times M_{t'}$:

$$\begin{aligned}
\left\| \widetilde{\boldsymbol{\mathcal{H}}}_i \right\| &\leq \sqrt{\left\| \widetilde{\boldsymbol{\mathcal{H}}}_i \right\|_{\mathrm{b},1} \left\| \widetilde{\boldsymbol{\mathcal{H}}}_i \right\|_{\mathrm{b},\infty}} \\
&\leq \left(\max_{\tau' \in [T], k \in \mathcal{N}} \sum_{\tau \in [T]} \left\| \widetilde{\boldsymbol{H}}_{k,\tau\tau',i} \right\| \right. \\
&\qquad \left. \cdot \max_{\tau \in [T], k \in \mathcal{N}} \sum_{\tau' \in [T]} \left\| \widetilde{\boldsymbol{H}}_{k,\tau\tau',i} \right\| \right)^{1/2} \\
&\leq \frac{\kappa}{2} T \left\| \widetilde{\boldsymbol{X}}_i \right\|.
\end{aligned}$$

This helps us establish an upper bound to the perturbation from the Hessian deviation:

$$\|E_i\| \leq \mathbb{E}\left[\left\| \widetilde{\boldsymbol{\mathcal{H}}}_i \right\| \left\| \widetilde{\boldsymbol{X}}_i \right\| \right] = \frac{\kappa T}{2} \mathbb{E}\left[\left\| \widetilde{\boldsymbol{X}}_i \right\|^2 \right].$$

The components of $\mathcal{E}_i$ can similarly be bounded:

$$\left\| \mathcal{E}_{k,i} \right\| \leq \|\mathcal{E}_i\| \leq \left\| \mathcal{J}\mathcal{V}^{\mathsf{H}} \right\| \|E_i\|$$

for both $k = 1, 2$. Note that since $(\mathbb{E}[\,\|a\|^2])^2 \leq \mathbb{E}[\,\|a\|^4]$ for any vector a, we also have

$$\left\| \mathcal{E}_{k,i} \right\| \leq \frac{\kappa^2 T^2 \left\| \mathcal{J}\mathcal{V}^{\mathsf{H}} \right\|^2}{4} \mathbb{E}\left[\left\| \widetilde{\boldsymbol{X}}_i \right\|^4 \right] =: r^2 \mathbb{E}\left[\left\| \widetilde{\boldsymbol{X}}_i \right\|^4 \right].$$

Combining all the results above, we can obtain the first moment stability. Even in this deterministic case, we still have

$$\begin{aligned}
\|I - \mathcal{M}_{11}\mathcal{D}_{11}\| &\leq 1 - \alpha\mu_{\max}, \\
\|\mathcal{D}_{12}\| &\leq \sigma_{12}, \\
\|\mathcal{D}_{21}\| &\leq \sigma_{21}, \\
\|\mathcal{D}_{22}\| &\leq \sigma_{22}.
\end{aligned}$$

However, the gradient noise terms are now replaced by the Hessian deviation terms. We then bound $\widetilde{Z}_{1,i}$ and $\widetilde{Z}_{2,i}$ by the same manner as before: via Jensen's inequality and set $t = \alpha\mu_{\max}$, we have

$$\begin{aligned}
&\left\| \widetilde{Z}_{1,i} \right\|^2 \\
&= \left\| (I - \mathcal{M}_{11}\mathcal{D}_{11}) \widetilde{Z}_{1,i-1} \right. \\
&\qquad \left. - \mathcal{M}_{11}\mathcal{D}_{12} \widetilde{Z}_{2,i-1} + \mathcal{M}_{11}\mathcal{E}_{1,i-1} \right\|^2 \\
&\leq \frac{\|I - \mathcal{M}_{11}\mathcal{D}_{11}\|^2}{1-t} \left\| \widetilde{Z}_{1,i-1} \right\|^2 \\
&\qquad + \frac{2}{t} \left(\|\mathcal{M}_{11}\mathcal{D}_{12}\|^2 \left\| \widetilde{Z}_{2,i-1} \right\|^2 + \|M_{11}\|^2 \left\| \mathcal{E}_{1,i-1} \right\|^2 \right) \\
&\leq (1 - \alpha\mu_{\max}) \left\| \widetilde{Z}_{1,i-1} \right\|^2 \\
&\qquad + \frac{2\mu_{\max}}{\alpha} \left(\sigma_{12}^2 \left\| \widetilde{Z}_{2,i-1} \right\|^2 + r^2 \mathbb{E}\left[\left\| \widetilde{\boldsymbol{X}}_{i-1} \right\|^4 \right] \right).
\end{aligned}$$

Similarly, set $t = 1 - \gamma$, we have

$$\begin{aligned}
&\left\| \widetilde{Z}_{2,i} \right\|^2 \\
&= \left\| -\mathcal{M}_{22}\mathcal{D}_{21} \widetilde{Z}_{1,i-1} + (\mathcal{J}_{22} - \mathcal{M}_{22}\mathcal{D}_{22}) \widetilde{Z}_{2,i-1} \right. \\
&\qquad \left. + \mathcal{M}_{22}\mathcal{E}_{2,i-1} + \mathcal{M}_{22}\mathcal{G}_{2,i}^{\star} \right\|^2 \\
&\leq \frac{\|\mathcal{J}_{22}\|^2}{t} \left\| \widetilde{Z}_{2,i-1} \right\|^2 + \frac{4}{1-t} \left(\|\mathcal{M}_{22}\mathcal{D}_{21}\|^2 \left\| \widetilde{Z}_{1,i-1} \right\|^2 \right. \\
&\qquad + \|\mathcal{M}_{22}\mathcal{D}_{22}\|^2 \left\| \widetilde{Z}_{2,i-1} \right\|^2 + \|\mathcal{M}_{22}\|^2 \left\| \mathcal{E}_{2,i-1} \right\|^2 \\
&\qquad \left. + \|\mathcal{M}_{22}\|^2 \left\| \mathcal{G}_{2,i}^{\star} \right\|^2 \right) \\
&\leq (1 - \gamma) \left\| \widetilde{Z}_{2,i-1} \right\|^2 + \frac{4\mu_{\max}^2}{\gamma} \left(\sigma_{21}^2 \left\| \widetilde{Z}_{1,i-1} \right\|^2 \right. \\
&\qquad \left. + \sigma_{22}^2 \left\| \widetilde{Z}_{2,i-1} \right\|^2 + r^2 \mathbb{E}\left[\left\| \widetilde{\boldsymbol{X}}_{i-1} \right\|^4 \right] + \left\| \mathcal{G}_{2,i}^{\star} \right\|^2 \right).
\end{aligned}$$

Combine the two recursions into one as follows:

$$\widetilde{\mathcal{Z}}_i^2 \leq \Gamma \widetilde{\mathcal{Z}}_{i-1}^2 + \begin{bmatrix} O(\mu_{\max}) \\ O(\mu_{\max}^2) \end{bmatrix} \mathbb{E}\left[\left\| \widetilde{\boldsymbol{X}}_{i-1} \right\|^4 \right] + \begin{bmatrix} 0 \\ O(\mu_{\max}^2) \end{bmatrix}$$

with $\widetilde{\mathcal{Z}}_i^2 := \mathrm{col}\{ \left\| \widetilde{Z}_{1,i} \right\|^2, \left\| \widetilde{Z}_{2,i} \right\|^2 \} \in \mathbb{R}^{2\times 1}$ and

$$\Gamma = \begin{bmatrix} 1 - O(\mu_{\max}) & O(\mu_{\max}) \\ O(\mu_{\max}^2) & 1 - \gamma + O(\mu_{\max}^2) \end{bmatrix}.$$

By the fourth moment stability, we conclude that

$$\begin{aligned}
\limsup_{i\to\infty} \widetilde{\mathcal{Z}}_i^2 &\leq (I - \Gamma)^{-1} \cdot \begin{bmatrix} O(\mu_{\max}^3) \\ O(\mu_{\max}^2) \end{bmatrix} \\
&= \begin{bmatrix} O(1/\mu_{\max}) & O(1) \\ O(\mu_{\max}) & O(1) \end{bmatrix} \begin{bmatrix} O(\mu_{\max}^3) \\ O(\mu_{\max}^2) \end{bmatrix} \\
&= \begin{bmatrix} O(\mu_{\max}^2) \\ O(\mu_{\max}^2) \end{bmatrix}.
\end{aligned}$$

We thus obtained the first moment stability:

$$\begin{aligned}
&\limsup_{i\to\infty} \left\| \widetilde{Z}_i \right\|^2 = O(\mu_{\max}^2) + O(\mu_{\max}^2) = O(\mu_{\max}^2) \\
\Rightarrow &\limsup_{i\to\infty} \left\| \mathbb{E}\left[\widetilde{\boldsymbol{X}}_i \right] \right\| = O(\mu_{\max}).
\end{aligned}$$